Assignment 2

Due: 30.10.2026

1. Introduction

In this assignment, you will debug some R functions for working with DNA sequences.

On ILIAS, you will find `assignment2_functions.R` and `assignment2_demo.R`. The functions file contains faulty code. The demo file runs small tests and shows the expected results.

2. The Functions

1. `gc_content(seq)` should return the percentage of bases in a DNA sequence that are G or C (e.g. `gc_content("GCGCATAT")` should return `50`).
2. `reverse_complement(seq)` should replace A with T, T with A, C with G and G with C, and then reverse the sequence (e.g.`reverse_complement("ATGC")` should return `"GCAT"`).
3. `transcribe(dna_seq)` should replace every T by U (e.g. `transcribe("ATGCGT")` should return `"AUGCGU"`.
4. `translate(rna_seq)` should translate an RNA sequence into amino acids, reading three bases (one codon) at a time. Stop and return the result immediately when you reach the stop codon `UAA`.

   Only the following six codons need to be supported:

   - `AUG` → `M`
   - `GCU` → `A`
   - `GGG` → `G`
   - `UUU` → `F`
   - `CCC` → `P`
   - `UAA` → Stop

   Example: `translate("AUGGCUUAAGGG")` should return `"MA"`.
5. `hardy_weinberg(p)` should return the genotype frequencies `AA` $= p^2$, `Aa` $= 2p(1-p)$ and `aa` $= (1-p)^2$. `hardy_weinberg(0.6)` should return `0.36`, `0.48` and `0.16`.
6. `random_dna(length)` should return a random DNA sequence of the requested length, using only A, T, C and G.

3. Your Task

1. Download both files from ILIAS and run `assignment2_demo.R`. Compare the actual outputs with the expected outputs.
2. Fix the bugs in `assignment2_functions.R`, one function at a time. Add comments above each corrected section explaining the bug. Keep the overall structure and test after every change.
3. Add one additional test case of your own for each function.